Expert Systems With Applications 252 (2024) 124148

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Review

Intelligent ship collision avoidance in maritime field: A bibliometric and systematic review

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ARTICLE INFO

Keywords:
Intelligent Ship
Collision Avoidance
Literature Review
Bibliometrics
Data Portrait

ABSTRACT

With the increasing promotion of digital technology, safety issues faced by rush ships in the maritime industry have once again received attention. Ship collision avoidance (SCA) is not only a hot topic in the shipping industry but also an eye-catching issue in the development of intelligent ships. This paper provides a bibliometric and systematic overview of the literature on SCA to assist researchers in understanding the frontiers and recent trends of SCA. Based on the bibliographic portrait, a classification and grading study was conducted on the literature to provide a systematic review of it. A screening process was conducted on 851 relevant articles related to SCA and published in 2004–2023 in the Web of Science (WoS) database, and 526 highly relevant and high-quality papers were selected. Then, CiteSpace, VOSviewer software, and data visualization techniques were used to conduct a bibliographic portrait analysis on the selected papers. The evidence from these systematic literature reviews revealed the close collaboration relationships among researchers, research institutions, and countries or regions in the field of SCA studying shortly. Furthermore, the frontiers of the research on SCA were focused on three aspects, i.e. research subjects, technological methods, and novel algorithms in intelligent SCA. COLREG is a problem that must be considered in intelligent SCA. The future trends in intelligent SCA were explored and SCA technology was introduced from artificial intelligence (AI) so as to meet the safety requirement of maritime autonomous surface ships. AI algorithms such as machine learning and deep learning are key technologies for SCA. This research provides a theoretical basis and implementation directions of the research on SCA. The hybrid encounters between traditional ships and intelligent ships, as well as multi-ship encounters in narrow waterways, will be the focal points of attention in the future.

1. Introduction

Maritime transportation plays a crucial role in the rapid development of economic globalization. According to statistics, 90 % of global trade is conducted through maritime transportation (UN, 2020). The global fleet saw a 3.2 % growth in 2022, and as of January 2023, 105,493 ships were weighing 100 tons or more, with a total tonnage of 2.27 billion deadweight tons (UNCTAD, 2023). Moreover, China is the largest commodity trading country and the second-largest economy in the world. It has completed approximately 95 % of its import and export

Abbreviations: ACO, Ant Colony Optimization; ARPA, Automatic Radar Plotting Aid; COLREG, Convention on the International Regulations for Preventing Collisions at Sea, 1972; CRI, Collision Risk Index; CTE, Cross-Track Error; DCPA, Distance to the Closest Point of Approach; FCS, Finite Control Set; GA, Genetic Algorithm; GAN, Generative Adversarial Network; IAPF, Improved Artificial Potential Field; LNDT, Local Normal Distribution-based Trajectory; IMO, International Maritime Organization; MANDS, Maritime Autonomous Navigation Decision System; MASS, Maritime Autonomous Surface Ship; MDP, Markov Decision Process; MPC, Model Predictive Control; NMPC, Nonlinear Model Predictive Control; OOW, Officers On Watch; PQSD, Potential Risk Quaternion Ship Domain; PRM, Probabilistic Roadmap Method; QSD, Quaternion Ship Domain; RL, Reinforcement Learning; RRT, and Rapidly-exploring Random Tree; SCA, Ship collision avoidance; SOLAS, International Convention for the Safety of Life at Sea; TCPA, Time to the Closest Point of Approach; USV, Unmanned Surface Vehicles; UUVs, Unmanned Underwater Vehicles; VO, Velocity Obstacle.

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https://doi.org/10.1016/j.eswa.2024.124148
Received 5 March 2024; Received in revised form 15 April 2024; Accepted 29 April 2024
Available online 2 May 2024

trade through shipping. Shanghai has become the largest container port in the world over the past few years (List, 2023). The enormous volume of transportation has made maritime transportation increasingly busy, particularly in restricted waters where the density of ships is higher. With the introduction of intelligent ships, SCA will become a core and critical issue in ensuring the safety of maritime transportation, as well as a difficult technical hurdle to overcome.

In the past, traditional ships have been the main focus of SCA research. The main objective of SCA in traditional ships has been to provide auxiliary driving and decision-making systems to relieve the workload of crew members and reduce human errors, thereby improving the safety performance of ships (Zhou, Daamen, Velling, & Hoogendoorn, 2019). In 2017, IMO first proposed the concept of autonomous ships and defined autonomous levels for autonomous ships in 2018. The goal of autonomous ships is to achieve fully automated navigation and SCA, which would significantly reduce human errors and accidents (Pietrzykowski, et al., 2022). MASS is categorized into four levels, which are ships with automated processes and decision support, remotely controlled ships with crew on board, remotely controlled ships without crew on board, and fully MASS. The introduction of MASS marks a shift from SCA research in traditional ships to SCA research in MASS. MASS aims to enhance maritime safety. They can make a series of navigation decisions instead of crew members, thereby overcoming errors caused by human factors and significantly reducing the probability of maritime accidents (Zhu, Lyu, Zhang, & Yin, 2022). Indeed, with the advancement of frontier technologies such as artificial intelligence, big data, and the Internet of Things, the world has entered the era of 'Industry 4.0' (Wang, Zhang, Zhang, & Gao, 2023). Traditional ships and MASS are now being covered by intelligent ship research. SCA technology is evolving from automation to intelligence, and intelligent SCA technology is triggering a technological revolution in the shipping industry, which has begun the era of intelligent ships. This intensifies the development of intelligent SCA research. Hence, for the foreseeable future, ships of different autonomy levels will coexist and share the responsibility of maritime transportation in a hybrid state. Moreover, there is currently a scarcity of research concerning SCA specifically focusing on different autonomy levels.

Literature research indicates that scholars have been striving to achieve intelligent SCA from various perspectives. The review and summary of research on intelligent SCA not only contribute to understanding the overall trend of this field but also provide new insights for achieving the highest level of autonomous ships and ensuring the navigational safety of intelligent ships in the future. Looking back at the research trajectory of intelligent SCA over the past few decades, during the period where traditional ships were taken as the focal point of investigation, the optimization and enhancement of SCA decision-making schemes from the perspective of SCA geometry were identified as the salient issues to be addressed. It is crucial to note that collision risk assessment, which includes collision probability, collision damage, and collision risk level, is one of the most critical issues in SCA decision support systems. Coric, Mandzuka, Gudelj, and Lusic (2021) provided a comprehensive review from a macro perspective on quantitative estimation models for ship collision frequency in 29 navigation channels and described their main modeling features. Ozturk and Cicek (2019) focused on the parameters and methods for individual collision risk assessment, integrating model-based analysis and parameterization clarification to provide inputs for navigation and maritime safety. Xiao, Ma, and Wu (2022) further elaborated on the dimensions of probability and consequence in risk analysis, dividing models for calculating ship collision probabilities and predicting outcomes into three categories: collision rate statistical models, statistical models, and simulation models. Meanwhile, the use of statistical methods based on historical accident data from accidentally has been considered to be the foundation of collision probability research. From an accidentology perspective, not only can the causes of collision accidents be analyzed, but collision risk can be quantified. Zhou, et al. (2019) reviewed the literature on maritime traffic models from the perspective of ship behavior modeling and found that ship behavior exhibited uncertainty, which could significantly impact the crew's judgment of collision risk. With the advent of MASS, humans have gradually been replaced by machines, leading to an increasing focus on the role of regulations in the SCA process. Wróbel, Gil, Huang, and Wawruch (2022) undertook a retrospective analysis of the incorporation of the COLREG in contemporary SCA algorithms. It was determined that additional attention to the status of COLREG in SCA was essential for the development of MASS. Lyu, et al. (2023) pointed out that designing a practical SCA decision-making algorithm has become a renewed challenge, owing to the complex constraints of the navigational environment, the influence of COLREG, and the underactuated characteristics of ships. The autonomous decision-making of the ship to avoid collision has been considered to be the core of the autonomous navigation of intelligent ships (Akdag, Solnor, & Johansen, 2022). In the era of intelligent ships, the SCA techniques of traditional ships and MASS have been covered by intelligent SCA technologies. Emerging intelligent technologies have garnered significant attention. The integration of intelligent ships with 5G technology has led to a proliferation of research on intelligent berthing, navigation, and control. SCA has become a focal point for scholars. Within the research on SCA, particular emphasis is placed on path planning and intelligent SCA decision-making. The problem of path planning is classified into global path planning and local path planning (Xu & Wang, 2014). Zhou, et al. (2020) summarized the research on multimodal constraint path planning into three stages based on its fundamental elements, such as shape, kinematics, dynamics, etc.: path planning, trajectory planning, and motion planning. It is generally recognized that intelligent SCA decision-making, which involves simulating the entire process of an excellent crew's manual SCA using computers, comprising the processes of perception, cognition, and decision formation, has gained significant recognition (He, et al., 2022; Jiang, An, Zhang, Wang, & Wang, 2022). However, a comprehensive systematic review of intelligent SCA has not yet been undertaken to date.

Due to the increasing number of research papers, the broadening perspectives, and the continuous iteration and update of research techniques, it becomes challenging to comprehensively understand the overall landscape of research on SCA through conventional literature reviews. For this reason, it is necessary to employ advanced technologies to organize the hot topics and trends related to SCA. Additionally, previous literature reviews in this field have primarily focused on the perception, cognition, and path-planning aspects of SCA, neglecting the systematic study of decision-making techniques in SCA. With the arrival of the big data era and the increasing dependency on new technologies, bibliometrics and visual analysis are considered the predominant methodologies for conducting literature reviews (Li, Meng, & Tong, 2021; Santos, Cavalcante, & Wu, 2023). Fu, Goerlandt, and Xi (2021) utilized the Visualization of Similarities viewer (VOSviewer) to identify the main thematic clusters in the field of Arctic risk management and found that the research primarily focuses on the strength of evidence in risk models, linearity versus complexity, and systematic accident theories. Cao, et al. (2023) in conjunction with Citation Space (CiteSpace) and VOSviewer, employed knowledge network mapping and cluster analysis to obtain knowledge network maps, research hotspots, research progress, and knowledge frontiers. The studies demonstrate that bibliometric analysis tools, such as CiteSpace and VOSviewer, are widely adopted by scholars in literature reviews and analyses, owing to their advantages in information visualization analysis and network topology analysis (Jiao, et al., 2021). Unfortunately, owing to the encapsulation of these two software tools in the Java programming environment, they are incapable of fulfilling certain specific knowledge graph and association analysis tasks. Supporting this notion, it is necessary to explore a new approach to literature analysis. The development of the Internet has led to an information explosion, necessitating the creation of a tool that provides precise information services, commonly known as a user portrait (Li, Ding, Xie, Gao, & Zhu, 2023; Ma, Kirilenko, &

Stepchenkova, 2020). The technique of literature portrait is proposed in this paper, which combines bibliometrics with big data analysis methods to conduct a comprehensive analysis of research hotspots and trends in the field of intelligent SCA, building upon the concept of user portrait.

The intellectual merits of this study can be summarized as follows:

(a) The technique of big data portrait is introduced in this study to analyze 526 literature works related to intelligent SCA, based on the CiteSpace and VOSviewer tools for information visualization. This work improves the analysis approach for literature reviews and enhances the capability for comprehensive and systematic network analysis on large-scale literature datasets.

(b) New research areas in intelligent SCA over the past two decades have been extensively explored in this study. Not only relevant topics in this research field are identified and analyzed, but the co-occurrence relationships among research hotspots, institutions, and authors are also precisely identified.

(c) Through bibliometric and profiling techniques, this study focuses on the research of decision-making technologies in intelligent SCA. It analyzes the research progress in this area, summarizes the research methods for collision risk, and delineates the development trends in the maritime field. Additionally, based on the analysis of bibliometric literature, we trace the knowledge frontiers of SCA development in the past 50 years, highlighting the frontiers and future directions of intelligent SCA in terms of collision objects, avoidance technologies, and avoidance algorithms.

In the following sections, several distinct components have been divided, each with its unique purpose and contribution. In particular, significant attention has been devoted to Section 2, which provided crucial insights into the sources and collection process of relevant literature, as well as the methodology employed to conduct the investigation. Moving forward, Section 3 served as a key component of the inquiry, with sophisticated analyses of bibliometric data and profiling information being offered to help unpack the numerous collaborative networks that emerged within the dataset. Building on this important foundation, Section 4 skillfully summarized and analyzed the breadth of research topics and methodologies underpinning the field of intelligent SCA. Meanwhile, Section 5 was specifically designed to offer fresh and innovative observations about the findings, effectively highlighting their significance in comparison to previous literature reviews and offering pragmatic suggestions regarding the future directions of research in this area. Ultimately, Section 6 brought out the conclusion.

2. Methodology and data

2.1. Methods

To illuminate the future development directions of research on intelligent SCA from a big data literature perspective, a research framework, as illustrated in Fig. 1, has been established, outlining the three phases of this study.

The first phase involved the creation of a literature database, encompassing the selection of keywords and the screening of relevant literature. The second phase has entailed literature profiling analysis, incorporating bibliometrics and data portrait. The specific approaches utilized have depended on the prominent themes that emerged. CiteSpace and VOSviewer were valuable tools that were used to analyze current trends and network knowledge maps within the specific field (Li, et al., 2021). CiteSpace has facilitated the highlighting of the core positions of keywords and their evolving trends over time, while VOSviewer's robust link and color functionalities have enhanced the representation of collaborative relationships among authors. Data portrait has enabled the spatial distribution of research hotspots to be highlighted, allowing for precise localization of research areas, institutional affiliations, and author relationships. The third phase involved a meticulous examination of 154 selected articles, building a classification repository for categorization. This facilitated the exploration of current research hotspots, key issues, and future development trends in the field based on the findings from literature portrait analysis.

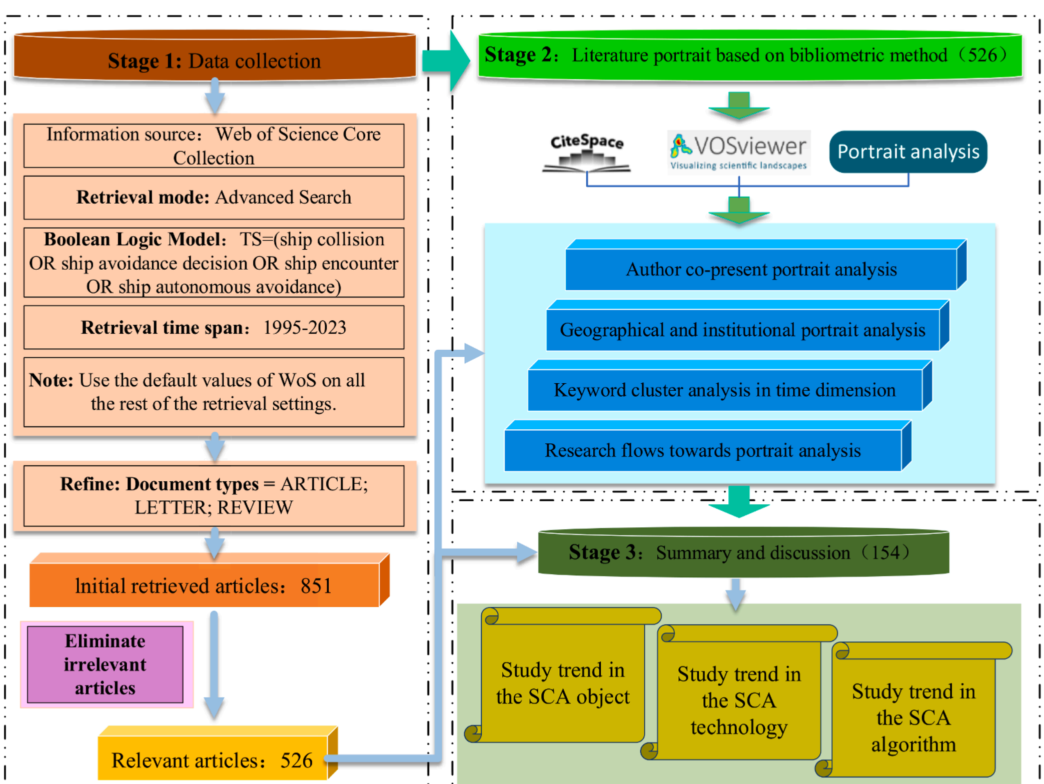

Fig. 1. Conceptual framework of the study.

2.2. Data

WoS is one of the most renowned scientific citation indexing databases in the world, encompassing over 12,000 high-impact journals (Cao et al., 2023). To ensure the authority and reliability of the research, the citation indexing databases (SCI-E and SSCI) were utilized by this study to retrieve original research papers and review articles related to SCA from the period of 2004 to 2023. The editorial and technical articles were excluded from the current search. Additionally, conference papers were not included as most conference papers are later published in higher-tier journals in a more comprehensive format. In literature metrics analysis and review, appropriate search criteria and strategies were chosen to ensure that the target research scope was fully covered. Intelligent SCA represents the highest level of autonomous SCA, which requires a focus on ship collision and avoidance principles. Therefore, "ship collision" was initially used as a search criterion. However, it was discovered that the number of articles was too large, and some articles did not pertain to collisions between ships, such as collisions between ships and whales or bridges. Some related articles, such as those on avoidance decisions and encounter situations, were also not included. As a result, the following strategies were eventually employed: "ship collision", "ship avoidance decision", "ship autonomous avoidance" and "ship encounter." To ensure timeliness and universality, the search time frame was set from "2004-01-01 to 2023-09-31", and the language of literature search was set to "English" In the end, 851 papers were obtained. Whereas the emphasis of these papers was on the discussion of SCA between ships, there were still a few papers studying collisions between ships and drilling platforms or marine life. Thereby, irrelevant papers were excluded through close reading, and finally, 526 papers related to intelligent SCA were obtained to support this study. The number of papers was counted by year, as shown in Fig. 2. From Fig. 2, it can be seen that the number of papers on intelligent SCA was low from 2004 to 2008. The number showed a slight increase from 2008 to 2017, which was closely related to the fact that the AIS device was mandated by the IMO to be installed on ships, providing a data basis for SCA research. With the rise of the IMO and artificial intelligence technology, MASS gradually entered the view of researchers from around the world. After 2018, the number of papers on intelligent SCA continued to rise and peaked at more than 120 papers in 2022. This indicates that the trend toward intelligent ships has become inevitable. Moreover, nearly half of the papers used in this study were sourced from "Ocean Engineering". Fig. 3 also indicates that this journal is highly regarded in intelligent SCA research. Following closely were "The Journal of Marine Science and Engineering", "The Journal of Navigation", and "Reliability Engineering & System Safety". This confirms the dominant position of "Ocean Engineering" in the field of intelligent SCA research.

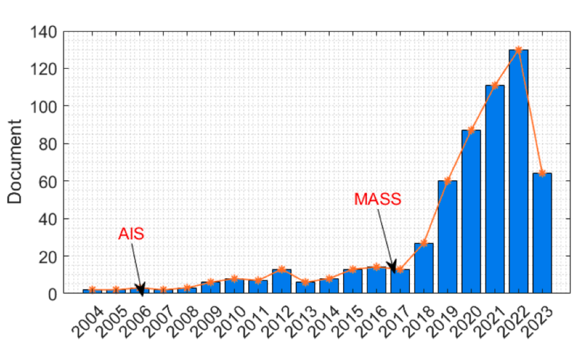

Fig. 2. Literature statistics from 2004 to 2023.

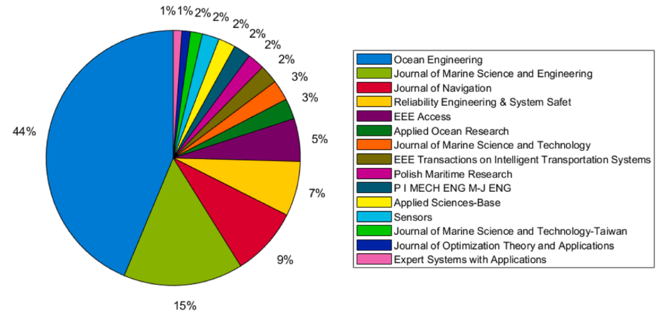

Fig. 3. Publications of relevant literature.

3. The portrait analysis of literature

3.1. The portrait analysis of geographical and institutional

The contribution of different countries to intelligent SCA research is represented in Fig. 4 using different colors. Specifically, the top ten contributing countries are selected and presented in Table 1. China is making significant contributions to global intelligent SCA research, ranking first with 98 published papers. Other major contributors include South Korea (48), Poland (41), and Norway (34). Given that China has significantly more publications in this research area than other countries, customized modifications are made to the color bar to provide a more nuanced representation of each country's contribution. Europe, being a longstanding maritime power, exerts a profound influence on SCA research, with Poland, Norway, and the UK serving as research centers and their surrounding countries as corresponding research bases. In East Asia, with China being the leading contributor, research centers are predominantly composed of China, Japan, and South Korea, with greater impact from China since the implementation of the Belt and Road Initiative. From an institutional perspective, the analysis shows that the top 10 universities in terms of publication output are Wuhan University of Technology (77), Dalian Maritime University (57), Hubei Key Lab Inland Shipping Technology (31), Shanghai Maritime University (30), Gdynia Maritime University (30), Delft University of Technology (28), Aalto University (28), Harbin Engineering University (20), University Lisbon (18), and Gdansk University Technology (17). Furthermore, the institutional cooperation network in the field of SCA is visible in Fig. 5. Based on the network analysis results conducted by VOSviewer, it is apparent that the various academic institutions closely collaborate in the field of intelligent SCA, forming a formidable scientific research network centered around Wuhan University of Technology.

Geographically, as shown in Fig. 6, most institutions are located in port cities. In China, they are predominantly concentrated in the Yangtze River Economic Belt, with Shanghai and Wuhan as the core, as well as in comprehensive waterway hubs centered around Dalian and Fujian. Wuhan University of Technology, Dalian Maritime University, and Shanghai Maritime University are focal points, radiating their influence throughout East and Southeast Asia. In Europe, the extensive coastline is dotted with numerous ports, and maritime-dependent European countries have dedicated significant efforts to intelligent SCA research. The cutting-edge research outcomes are primarily concentrated in three institutions: Gdynia Maritime University, Delft University of Technology, and Aalto University. Overall, countries that rely on or emphasize ocean development demonstrate the highest productivity in research. It is worth noting that since China proposed the Belt and Road Initiative, advocating for win–win cooperation among nations, the country has vigorously promoted independent research and development, enabling the transformation from "Made in China" to "Created in China". Thus, China's intelligent SCA research has been catching up with that of Europe, the United States, Japan, and South Korea. Leveraging the advantages of prosperous ports, various institutions have prioritized the development of marine-oriented specialties, making

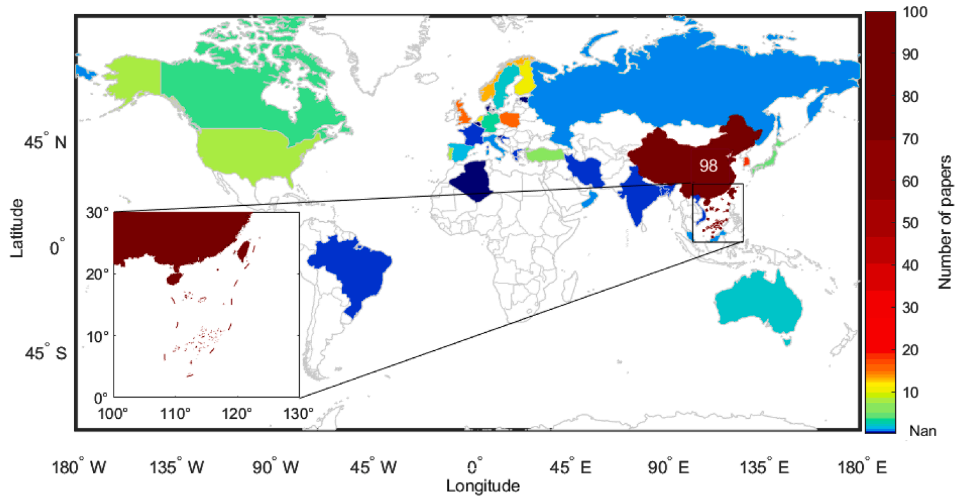

Fig. 4. The Portrait Analysis of Geographical.

Table 1
Top 10 countries publishing papers related to SCA.

No.	Country	Document
1	PEOPLES R CHINA	98
2	SOUTH KOREA	48
3	POLAND	41
4	NORWAY	34
5	ENGLAND	30
6	USA	26
7	FINLAND	26
8	NETHERLANDS	26
9	PORTUGAL	20
10	TURKEY	14

outstanding contributions to intelligent SCA research. In 2017, “Da Zhi”, the world’s first intelligent ship certified by a ship classification society, was successfully developed in China. In 2022, the world’s first unmanned autonomous ship successfully crossed the Atlantic. On October 18, 2023, “Zhi Fei”, China’s first intelligent container ship, achieved autonomous SCA by relying on an intelligent navigation system under the remote control of onshore operators for berthing, departure, and remote piloting. This marks the era of intelligent ships, as the highest level of MASS has been attained.

3.2. The network and portrait analysis of Co-authorship

A total of 526 papers are used to analyze the collaboration among authors, which can reveal the collaboration patterns of researchers worldwide. VOSviewer is employed in this study to visualize the collaboration network among authors. Authors participating in the analysis have contributed at least two papers. The visualization network is comprehensively and systematically constructed to gain insights into the collaboration among authors in the field of SCA. In the collaboration network graph, the size of nodes represents the number of papers published by authors, the thickness of connecting lines indicates the strength of the connections between authors, and the colors of nodes and lines represent the nature of collaboration among authors. Data portrait is utilized for analyzing the intensity of collaboration among teams. 15 different collaborative teams in SCA research are demonstrated in Fig. 7. Team T4 led by Yamin Huang and Team T2 led by Jinfen Zhang are the

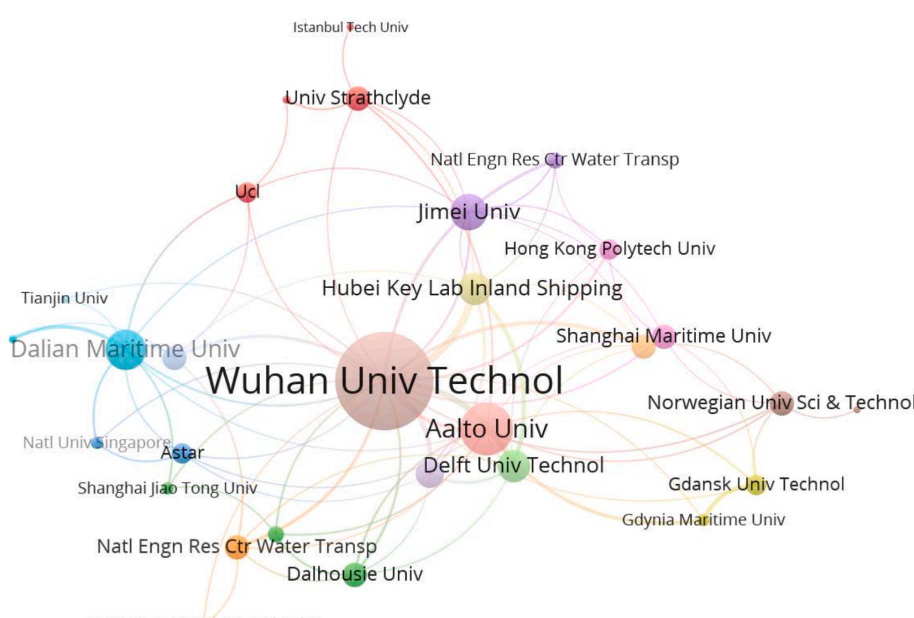

Fig. 5. Institutional cooperation network in the field of SCA.

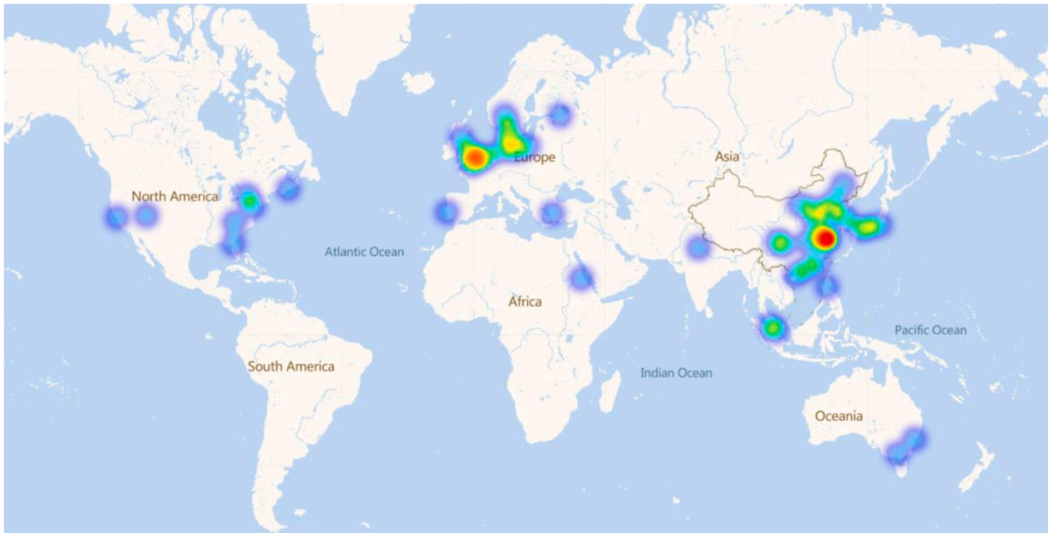

Fig. 6. Spatial distribution of institutional.

top two contributors. It is worth noting that most of the researchers in these two teams are affiliated with Wuhan University of Technology, while a portion of them come from Dalian Maritime University, Shanghai Maritime University, Gdynia Maritime University, and Delft University of Technology. This further verifies the prominent position of Wuhan University of Technology in the field of SCA research. Nevertheless, some prominent researchers, such as from the research groups at MIT (Michael Benjamin), Denmark (Mogens Blanke and Roberto Galeazzi), and Norway (A. Lekkas, T. A. Johansen, A. Rasheed), may not appear in the illustration of results due to statistical reasons. They have conducted detailed studies on regulations for achieving collision avoidance in the maritime sector, and their research findings have had a profound impact on the study of autonomous ships (Eriksen, Bitar, Breivik, & Lekkas, 2019; Hansen, Enevoldsen, Papageorgiou, & Blanke, 2022; Randeni, Sacarny, Benjamin, & Triantafyllou, 2022).

3.3. Research hotspot and issue frontier

CiteSpace not only can identify and extract relevant themes, but it can also cluster research topics based on their correlations over time (Fu, et al., 2021). To highlight the research hotspots more intuitively, parameters were automatically set to hide those appearing less than 10 times. As shown in Fig. 8, the generated cluster labels represent corresponding research topics, with each node in the timeline representing a different research topic. Among all the topics, the highly cited keywords in each cluster label represent the key content of that topic. The analytical results horizontally illustrate the evolution of research hotspots over time from the temporal dimension, and vertically indicate the popularity of each research topic from the size of the keyword cluster family.

SCA is a T0-level vocabulary item with a research heat dating back to 2004. It represents a core topic in the field of intelligent SCA and is closely related to almost all subsequent hot topics. Following closely behind are deep learning, decision support, and artificial potential field

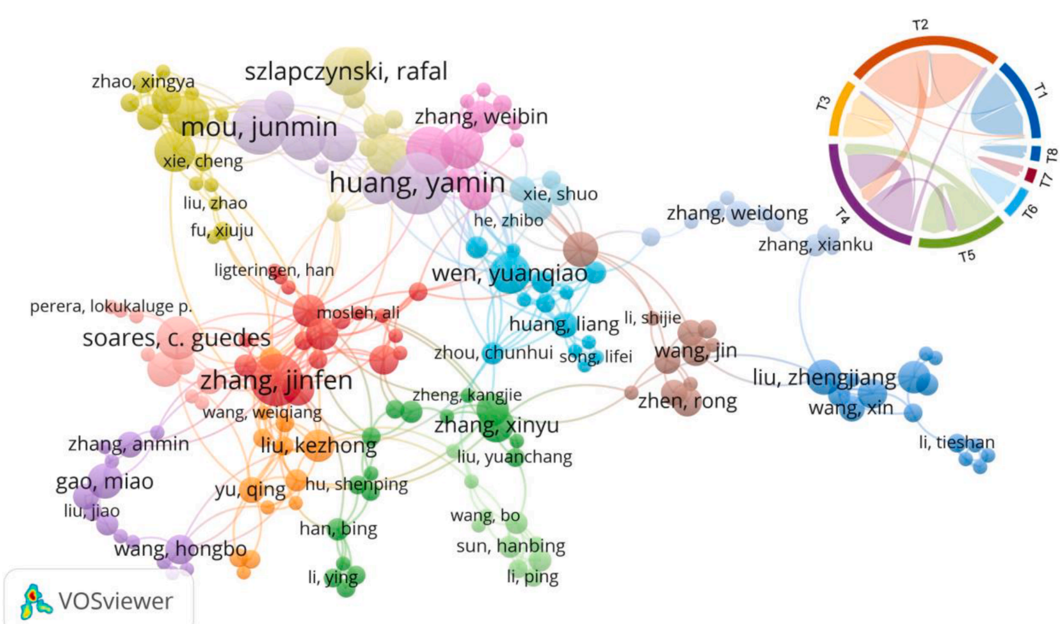

Fig. 7. The analysis of Co-authorship network.

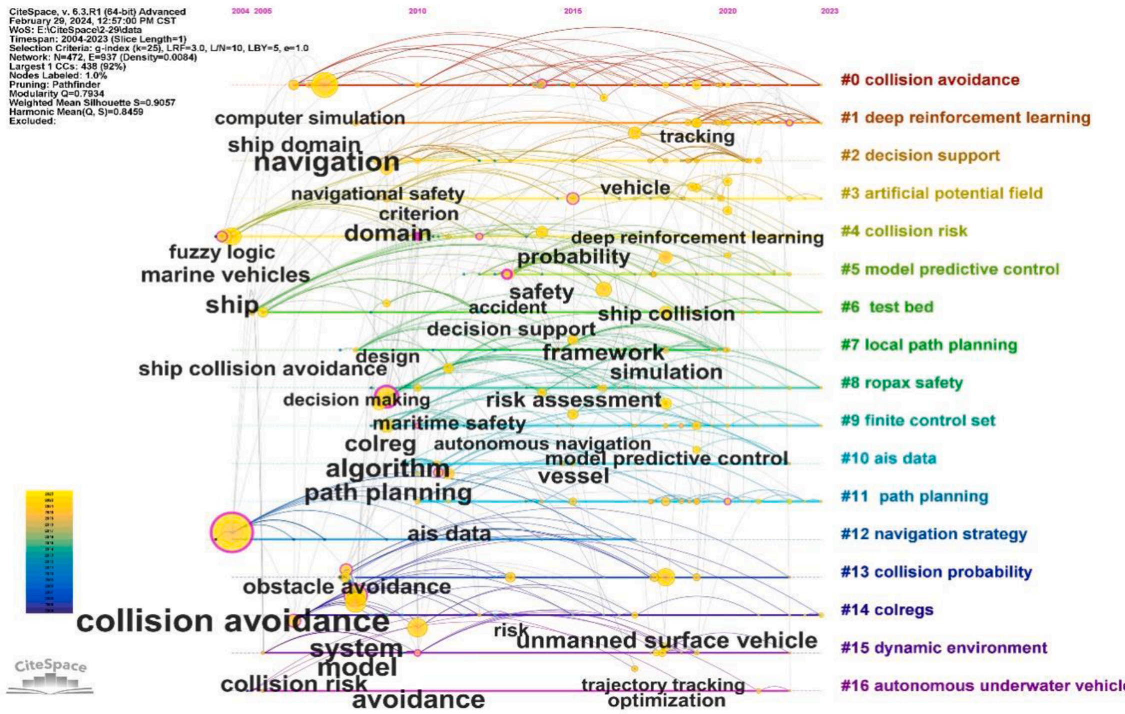

Fig. 8. The timeline results of keywords on SCA after clustering.

(APF) algorithms. Among them, the APF is one of the most classic path-planning algorithms in the field of intelligent SCA and has been used in research for a long time. Deep learning is a new research direction in the field of machine learning, and its emergence brings us closer to the highest goal of MASS, which is a fully autonomous ship. Decision support is the core of intelligent SCA, and it determines the degree of humanization of intelligent SCA. Furthermore, the analysis shows that AIS data and COLREG also play important roles in the research of intelligent SCA. This indicates that AIS data will remain fundamental research data in the past and future. There has been intense debate in the field of SCA research concerning whether intelligent ships will need to adhere to COLREG in the future (He, et al., 2017; Zhang, Zhang, Yan, Haugen, & Soares, 2015). Simultaneously, MPC should not be disregarded as it encompasses several interdisciplinary fields including control engineering and mathematical modeling. This lends complexity to the research. However, it should receive adequate attention as a crucial step from theory to practice in intelligent SCA. It is worth noting that CiteSpace may ambiguously truncate certain keywords, partly due to potential inclusion or containment relationships. For instance, this includes the relationship between local path planning and overall path planning, as well as the connection between MPC and FCS. Nevertheless, we believe this is necessary. On one hand, it underscores the significance of local path planning within the broader context of path planning. On the other, it reveals the developmental trends of MPC technology (Martinsen, Lekkas, & Gros, 2022). The combination of FCS and MPC optimizes the target optimization and switching state decision-making processes into a single step. This approach offers advantages such as simple concepts, wide applicability, and the ability to easily incorporate constraints and nonlinearity into value functions. Consequently, it has emerged as one of the research hotspots in recent years. Collision risk and ship safety represent focal points in the investigation of intelligent SCA, specifically in the perception and cognition stages of SCA. The former encompasses the perception of physical collision risk as well as system safety collision risk, while the latter concentrates on risk perception in the algorithm implementation process for MASS. Finally, thanks to the dedicated efforts of researchers worldwide, the technology of autonomous underwater vehicles has reached a relatively mature stage, providing valuable insights and experiences for MASS research.

The above-mentioned study provides a detailed analysis of the regions, institutions, authors, and keywords in the field of intelligent SCA. This research approach is also widely recognized within the academic community. However, this modular analysis emphasizes the characteristics within each module while neglecting the connections between modules. Therefore, it is necessary to further investigate the research context through the use of data profiling techniques. The Sankey diagram in Fig. 9 illustrates the flow relationships between research institutions, authors, and research hotspots. It is evident that Wuhan University of Technology has contributed a significant number of researchers and has shown interest in various domains of intelligent SCA. Other institutions and researchers also have their areas of focus. For example, the team led by Kujala, P from Aalto University has shown a high dedication to SCA, COLREG, and deep reinforcement learning (Zhang, Kujala, Musharraf, Zhang, & Hirdaris, 2023). Furthermore, many researchers are affiliated with multiple institutions, which greatly facilitates the exchange of cutting-edge issues between different disciplines. This collaborative mindset is beneficial rather than detrimental to the study of intelligent SCA.

Overall, the research hotspots and knowledge frontiers of intelligent SCA show that research questions and methods coexist. In terms of research questions, the study addresses various aspects through the implementation of the perception, cognition, decision-making, and control modules. Methodologically, based on the principles of SCA, intelligent algorithms, and biomimicry are leveraged to optimize decision-making for intelligent SCA and strive to address control issues in MASS from a multidisciplinary perspective. The evolution of research in intelligent SCA is analyzed using data profiling techniques. The research reveals that it has gradually formed a scientific research system involving multiple problems and interdisciplinary methods. Both research institutions and researchers have their research characteristics, and they are actively contributing to the forefront of intelligent SCA.

4. The topic of intelligent SCA

Based on bibliometric analysis and data portrait techniques, it can be

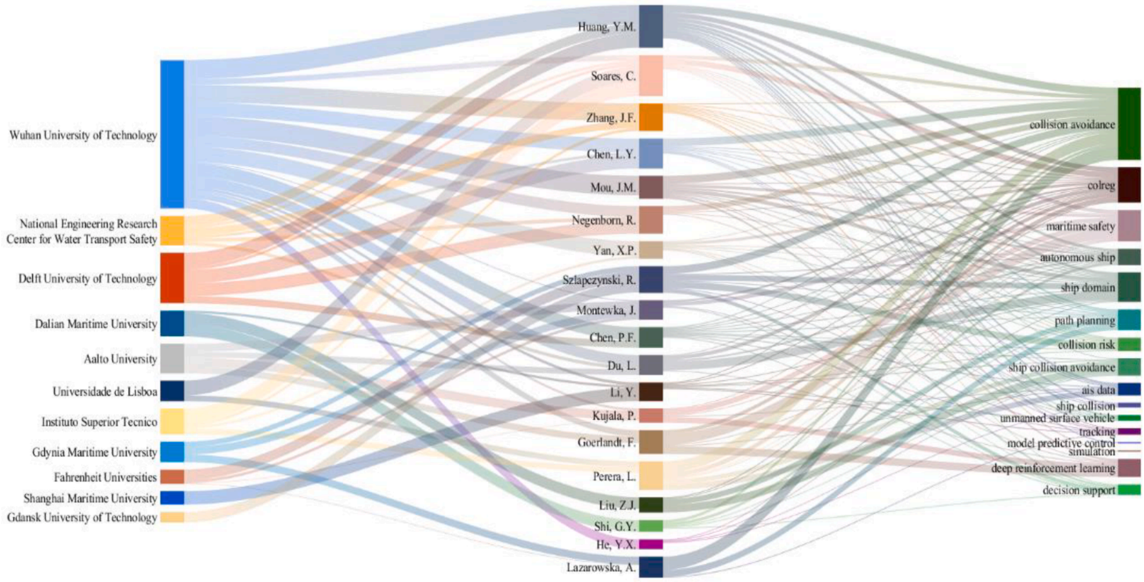

Fig. 9. The flow direction of research hotspot portrait analysis.

observed that intelligent SCA is a broad and interdisciplinary research area with many focus points. Fig. 10 presents the scenario where an intelligent ship encounters both MASS and a Traditional ship on the sea surface in the era of intelligent navigation. The predictable point of collision (PPC) and predictable area of danger (PAD) formed by the intelligent ship and the autonomous ship depicted in the diagram clearly illustrate the essence of ship collision, which occurs when two ships in a meeting situation simultaneously arrive at the same location (Xin, Yang, Liu, Zhang, & Wu, 2023). In radar SCA, the predicted dangerous "point" is expanded into a predicted dangerous "area". PAD refers to the area that the own ship and the target ship may enter at some future time. To avoid this area, the own ship maintains a certain safe encounter distance with the target ship and continues sailing along the direction indicated by the vector, thus entering the PAD. At this point, the target ship will enter the area required for the own ship's safe navigation, posing a threat to the own ship's safety. The typical method to avoid danger is to maneuver the ship to rotate counterclockwise with sufficient speed. In practical SCA, the SCA actions taken should comply with the requirements of COLREG. In the era of intelligent SCA, intelligent ships must comply with COLREG to safely give way to target ships. Therefore, it is essential to fully consider COLREG in the research of intelligent SCA (Xie, Gang, Zhang, Liu, & Lan, 2024). The intelligent ship and the MASS in the diagram are in a situation of crossing encounter. According to the requirements of the COLREG, the intelligent ship is a give-way ship and needs to turn to starboard to give way (Zhou, et al., 2024).

With the advent of the information age, intelligent ship systems have evolved from single-ship intelligence to a new form of intelligent shipping based on ship-shore coordination. Ship remote control technology is one of the forms of ship-shore coordination. Its essence lies in the shore-based manual driving of the ship, achieving efficient transmission of ship-to-shore navigation information, three-dimensional reliable presentation of ship-to-shore control center's navigation information, real-time reliable transmission of shore-based remote control commands, and reliable execution of shipborne systems through online monitoring systems, shore-based telemetry and remote control systems, and shipborne response systems (Wang, Zhang, Liu, Wang, & Zou,

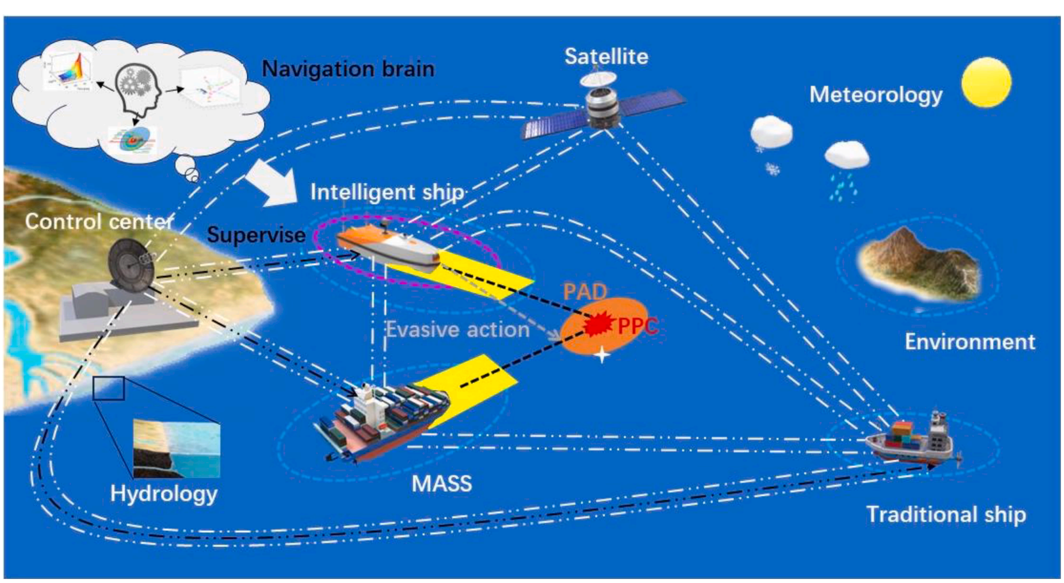

Fig. 10. Response of intelligent SCA.

2023). In ship-shore coordination, intelligent ships can assess collision risks based on perceived geographical, meteorological, hydrological, surrounding geographical environment, and target ship conditions. The shore control center can also monitor whether there is a collision risk by analyzing and processing integrated data. The core of ship-shore coordination is to establish an integrated and systematic decision-making system to optimize the allocation and balance of ship-shore information resources. It emphasizes real-time interaction and dynamic adjustment between shore and ship, as well as rational division of labor and coordinated control between ship and shore. Additionally, satellite-based coastal systems are required to assist in generating optimal SCA decision-making within the ship's navigation systems. Researchers in this field are currently focusing on realizing this process.

This section aims to provide an overview of the research frontier and developmental trends of intelligent SCA from three perspectives: research subjects, SCA technologies, and SCA algorithms.

4.1. Objects of SCA research

In terms of the research focus, the development of SCA has undergone three successive stages, namely, the traditional ship development stage, the autonomous ship evolution stage, and the intelligent ship evolution stage. In the traditional ship development stage, the research primarily focused on the issues of supporting senior deck officers in making SCA decisions based on geometric principles. During the long period of navigation practice, OOW made SCA decisions based on the factors encompassing encounter situations, COLREG, and professional knowledge (Li, Mou, Chen, Huang, & Chen, 2021). In this process, CPA was often employed to assess the level of danger posed by the encountering ships. Furthermore, DCPA and TCPA are typically used together to determine the appropriate timing of SCA. Representative approaches included the development of an intelligent SCA decision model based on DCPA and TCPA using fuzzy decision theory, which offered action plans based on the timing of evasive maneuvers (Shi, Zhen, & Liu, 2022). Due to its visibility, radar SCA has become an indispensable means. As early as the early 20th century, the International Maritime Organization (IMO) issued the SOLAS, requiring all ships with a gross tonnage of 10,000 or above, built after July 1, 2002, to be equipped with ARPA devices. Since then, the functionality of SCA systems on ships has been primarily achieved through ARPA. Intelligent SCA based on ARPA allows the simultaneous estimation of the target ship's motion state (i.e., position, heading, and speed) and geometric parameters (length) within the framework of extended Kalman filtering (Ozoga & Montewka, 2018). During the evolutionary phase of MASS, a clear direction for autonomous SCA has emerged, due to the IMO has established definition of MASS. The objective is to reduce the contribution of human factors and enhance navigation intelligence, reliability, and safety. MASS can utilize perceptual information to replace OOW, and provide optimized navigation decision-making solutions through expert knowledge bases and neural network systems. Wang, Zhang, Zhang, et al. (2023) proposed an elaborately designed architecture for MANDS, which incorporates appropriate decision models and communication interfaces with external systems. The intelligent ship evolution stage also includes research on intelligent SCA for UUVs (Randeni, et al., 2022). Additionally, the advanced version of MASS can be considered a research focus for SCA during the intelligent ship evolution stage. Compared with MASS, USVs exhibit better maneuverability and control performance (Landsburg, et al., 2005). Currently, intelligent SCA theories are typically applied to small and medium-sized ships ranging from 2 m to 15 m in length (Qdven, Martinsen, & Lekkas, 2022; Sun, Wang, Fan, Mu, & Qiu, 2018). As a smaller intelligent ship, USVs are increasingly favored by researchers.

4.2. SCA research on technical perspectives

Research has shown that the steps involved in SCA for human navigation and intelligent navigation are homologous. The intelligent decision-making layer of a MASS plays a "co-pilot" role in the entire unmanned driving system (Wang, Zhang, Cong, Li, & Zhang, 2019). In light of this, research on SCA decision-making holds significant importance and occupies a considerable portion of the field of SCA. This view is validated through keyword analysis, in the "decision support" section. SCA decision-making is defined as the action selected by a ship within the range of possible actions when collision risk between the ship and the target ship reaches a certain degree (Wang, Zhang, & Li, 2020). Therefore, the degree of collision risk has always been a hot topic in research on SCA, as it is a critical factor in decision-making. Moreover, from the definition of SCA decision-making, it is apparent that avoidance behavior is also a critical factor. However, the timing of taking action and the safe passing distance can be described by the ship domain (Xiao, Ligteringen, van Gulijk, & Ale, 2015). As such, collision risk, ship behavior, and decision-making technologies are regarded as the three crucial components of research on intelligent SCA in this study.

1) Research Developments in Ship Risk of Collision

As it is well known, collision risk identification is a necessary condition for intelligent ships to recognize risks and make correct SCA decisions in different situations. Within the framework of SCA rules, collision risk determines the manner and timing of evasive actions. Consequently, collision risk in maritime navigation has always been a significant topic in the research of intelligent SCA. Over the years, there have been different perspectives on the definition of collision risk. Initially, scholars generally regarded the meaning of collision risk as "possibility" or "probability". Nevertheless, with the advent of the big data era, the use of ship domain to describe collision risk has become increasingly reliable (Weng, Du, Shi, & Liao, 2023). The current ship domain is not only a big data problem but also an uncertainty problem. Existing research has demonstrated that probability density can describe the probability of ship collisions. For example, Im and Luong (2019) used Gaussian functions to construct the ship domain to obtain continuous values of collision risk. Zhu, Xi, Hu, and Chen (2023) employed Gaussian mixture clustering to obtain the uncertain range of ship motion. In mathematics, probability refers to the likelihood of an event occurring in a specific environment, i.e., predicting the likelihood of an event happening based on parameters corresponding to the environment before the outcome occurs. Conversely, likelihood is the inference of the possible environment that may produce a specific result under a specific outcome. As mentioned earlier in the context of Deep Reinforcement Learning (DRL), the realization of intelligent ships involves learning and memorizing historical data, with the key component of value reward utilizing the likelihood function (Rawson & Brito, 2023). This indicates that in the research of intelligent SCA, from a mathematical logic perspective, collision risk should be understood as "likelihood". Many scholars have already used "likelihood" to describe collision risk, although they primarily focus on studying the "chance" and "consequences" of ship collision accidents from an accidentology standpoint (Rawson & Brito, 2021). Nevertheless, this also demonstrates the legitimacy of using "likelihood" to describe collision risk.

In the field of risk quantification, scholars have discussed collision risk from the perspectives of macro-collision risk and micro-collision risk (Chen, Huang, Mou, & van Gelder, 2019). The former focuses on the probability of collisions in a certain area at a specific time (Altan & Otay, 2018), while the latter examines collision risk between two ships during an encounter (Ståhlberg, Goerlandt, Ehlers, & Kujala, 2013). In intelligent SCA, micro-collision risk is the most relevant factor. To quantify collision risk as a numerical value, IMO has defined the CRI. Previous studies have found that factors such as DCPA, TCPA, relative distance, relative bearing, and relative velocity have a significant impact on ship collision risk (Zhao, Li, & Shi, 2016; Zhen, Shi, Liu, & Shao, 2022). Whereas, there is considerable ambiguity and uncertainty among these factors. Strictly speaking, ship attributes should not be neglected in the quantification of collision risk. Hitherto, the best way to eliminate the influence of ship property is to use the ship's length to nondimensionalize the DCPA and TCPA (Szlapczynski, 2006; Zhao & Fu, 2021). The study of CRI has played a significant role in defining the four stages of ship collision danger. Both Zhu, Xi, Hu, Wu, and Han (2023) and Im and Luong (2019) have given new advice on collision risk versus avoidance timing from a dynamic collision risk perspective.

2) Research status of the ship domain

The concept of ship domain is crucial in the study of intelligent SCA, as it describes the timing of evasive actions and the safe passing distance (Liu, Shi, & Zhu, 2022). Thus, the ship domain can be understood as the water area required for safe and effective ship navigation at any given time. Scholars have made significant efforts to define the effective domain. Over the past few decades, three methods have been used to formalize a conceptual ship domain, as shown in Fig. 11. One method is the mathematical modeling approach based on ship maneuvering equations. For example, Fujii and Tanaka (1971) elliptical ship domain, Goodwin (1975) combined-sector ship domain (Hörteborn, Ringsberg, Svanberg, & Holm, 2019), circular domain, polygon domain (Pietrzykowski & Uriasz, 2009), and QSD (Wang, 2013). The original ship domains defined by Fujii and Goodwin vary with the length of the ship. On the sea, the relationship between a ship and its length is intrinsic. Therefore, ship behavior often reflects the behavior of the crew. Goodwin hypothesized that the ship domain on the starboard side is greater than that on the port side. Wang (2013) believed that the size of the domain is determined by a quaternion comprising four radii, including the bow, stern, starboard, and port side After considering factors that influence the domain (i.e., ship maneuverability, speed, heading, etc.) and COLREG, QSD was proposed. The introduction of QSD made ship domains more flexible and reliable. Subsequently, Wang (2012) developed Fuzzy QSD by considering uncertainty and fuzzy information. With the popularization of AIS devices and the disclosure of AIS data, the era of big data for SCA has arrived. Scholars have paid attention to a data-fitting method based on the context of big data (Li, Li, Qi, Li, & Wu, 2024). Over the past decade, AIS data has been used to demonstrate the existence of ship domains and to verify different shapes of ship domains, such as elliptical ship domains and QSD (Breithaupt, Bensi, & Copping, 2022; Weng et al., 2023). To adhere to the "early, large, wide, clear" SCA principle, the concept of dynamic boundary for the ship domain is introduced. Its existence avoids the formation of urgent situations during ship encounters and ensures a sufficient safe navigation distance. This concept is similar to that of the potential ship domain. However, an early shortcoming of ship domain research was the inability to describe the degree of danger at any point around the ship. To address this, Im and Luong (2019) proposed the Potential QSD model, which has a clear indication of risk level. The third and most cutting-edge approach is the

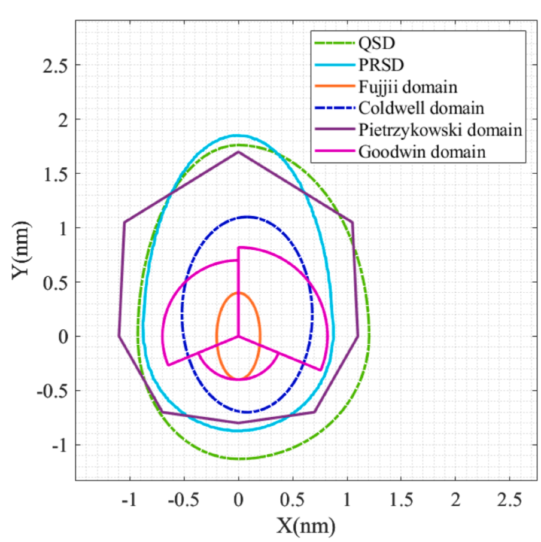

Fig. 11. Comparison of different ship domains.

dynamic ship domain considering the uncertainty of ship motion. Qiao, Zhang, and Wang (2021) considered the uncertainty of ship position prediction and proposed an improved QSD integrating field theory. The uncertain ship motion in the big data scenario is represented by a joint probability distribution. Therefore, using a joint probability density function to describe the ship domain can provide not only the bounds and boundaries of the ship domain but also the comprehensive collision probability (Zhu, et al., 2023). Ship domain plays a critical role in the current stage of intelligent ship path planning, and scholars often use the concept of ship domain to assess the level of collision risk and determine decision-making strategies.

3) Current situation of intelligent SCA decision-making technology

Autonomous SCA decision-making is one of the most important issues for IMO, governments, academic institutions, and scholars. The research on autonomous decision-making has gone through three stages. The first stage is the auxiliary decision-making stage based on SCA geometry. SCA of ships is often regarded as two points moving at a certain velocity vector. Describing this motion state based on a vector triangle forms the basis of geometric SCA. In SCA decision-making, there are two ways to detect the need to take evasive action. On the one hand, the current dangerous status serves as the basis for evasion. DCPA and TCPA are considered key factors in determining the formation of danger in SCA decision-making (Zhen, Riveiro, & Jin, 2017). On the other hand, predicting danger can also serve as the basis for evasion. The possibility of mutual collision between two ships in the same area always exists (Szlapczynski & Szlapczynska, 2017). Therefore, possible collision points and possible collision areas can also serve as the basis for SCA decision-making (Tsou, 2016). In addition, decision-making based on a geometric model can also calculate CTE and heading errors in the guidance process (Sivaraj, Dubey, & Rajendran, 2023). The output of SCA decision-making is the ship's evasion behavior. Typically, ship evasion behavior includes the timing, motivation, mode, magnitude, and effectiveness of avoidance. The second stage is the stage of autonomous SCA decision-making considering COLREG. Research has found that not following the rules and making incorrect avoidance maneuvers can lead to chaos and potentially catastrophic collisions. Statistics show that 60 % of maritime casualties are caused by collisions, and 56 % of collisions are caused by violations of COLREG (He, et al., 2017). Therefore, SCA must comply with COLREG, and intelligent ships should not hinder the navigation of traditional ships. Zhang, et al. (2015) proposed a distributed multi-ship SCA decision-making approach that considered rules. In situations involving multiple ship encounters, the violation of COLREG by one ship sometimes increases the difficulty for other ships to avoid collision and could cause other ships to take unnecessary control measures (Woerner, Benjamin, Novitzky, & Leonard, 2019). Therefore, intelligent SCA decision-making should comply with the requirements of COLREG. Li, Wang, Wu, and Ni (2023) proposed a more advanced approach, which was a SCA decision-making support scheme that considers ship maneuverability and COLREG. It combines the IAPF method and the NMPC algorithm. However, the difficulty of incorporating COLREG into navigation technology lies in the fact that, unlike land-based navigation, the rules encountered by ships are unique, especially as each encountered scenario is different. The third stage is the intelligent SCA decision-making stage based on big data. Intelligent SCA in the big data environment exhibits two problems, one is the complexity of the ship's motion environment, and the other is the uncertainty of the ship's behavior pattern (Wang, Zhang, & Zheng, 2021). Regarding the former, real-time collision risk can be obtained from the perspective of multi-source information fusion to provide a basis for decision-making. Regarding the latter, sufficient empirical data can be trained using deep learning to make the best decision-making plan (Deraj, Kumar, Alam, & Somayajula, 2023). Jiang, et al. (2022) considered both collision risk and decision-making in their research on humanoid collision methods. The current autonomous decision-making is defined as four stages: observation, inference, prediction, and decision (Wang, Wu, Zhang, Wu, & Wang, 2020).

4.3. Evolution of the SCA algorithms

The intelligent SCA algorithm can be broadly divided into three stages: the mathematical model stage, the intelligent algorithm stage, and the big data-based machine learning stage. In the early stages of SCA research, the main focus was on establishing mathematical models to quantify collision risks. There are typically three approaches used. One approach is based on indicator reasoning methods, such as neural networks (Shi, Zhang, & Agarwal, 2015), fuzzy theory (Chen, Ahmad, Lee, & Kim, 2014), support vector machines (Park & Jeong, 2021), and cloud models (Liu, Zhang, & Liu, 2020). Another approach involves fitting mathematical formulas, as proposed by Mou, van der Tak, and Ligteringen (2010) who developed a formula considering the negative exponential to directly obtain collision risk from DCPA and TCPA. Similar approaches can be found in Table 2. Studies have shown that the mathematical formula fitting method has the advantages of fast computation and accurate fitting results in a big data environment. Apart from these two approaches, the ship domain can also be used to detect collision risks. Liu, et al. (2022) proposed a ship collision risk assessment model based on Fuzzy QSD. This model can provide continuous and intuitive measures of collision risk.

The intelligent SCA algorithm stage included classical intelligent algorithms and bio-inspired intelligent algorithms. The A* algorithm demonstrated a good performance in global path planning problems (Ma, Chen, Huang, Yan, & Wang, 2016; Singh, Sharma, Sutton, Hatton, & Khan, 2018). It was the most representative of classical intelligence algorithms. However, SCA decision-making techniques need to consider static and dynamic obstacle problems. The APF algorithm has received widespread attention for its ability to ship static relationships between ships and effectively avoid dynamic obstacles (Yang, Wang, Wu, & Wang, 2011). Unfortunately, both of these methods were unable to solve the problem of local path planning, particularly APF's tendency to fall into local minima. Given this, Zhang, et al. (2015) proposed an improved VO method based on the typical VO method and the CPA model to address the local path planning issue. Furthermore, the IAPF algorithm also possesses the capability of local path planning (Wang, et al., 2020). The VO method was relatively easy to achieve SCA goals while adhering to COLREG (Statheros, Howells, & McDonald-Maier, 2008). Moreover, the VO algorithm aligns well with the principles of SCA geometry and was an important method in the second generation of intelligent SCA decision-making. Researchers have developed various improved VO algorithms to address different issues encountered during application. For instance, Non-Linear Velocity Obstacle (NLVO) (Chen, Huang, Papadimitriou, Mou, & van Gelder, 2020), Probabilistic Velocity Obstacle (PVO) (Li, et al., 2022), Reciprocal Velocity Obstacle (RVO) (Xue, Wu, Yamashita, & Li, 2023), and Generalized Reciprocal Velocity Obstacle (GRVO) (Wang, Zhang, Song, & Mao, 2023). However, SCA is a highly complex process, and it is challenging to accurately capture this process. Fortunately, the long history of navigation has accumulated valuable knowledge and experience for safe SCA. This knowledge and experience provide the prerequisites for introducing expert systems into intelligent SCA decision-making. Expert systems simulate the decision-making process of human experts to solve complex problems that require advanced seafarer handling (Bakdi & Vanem, 2022). Bio-inspired intelligent algorithms not only enhanced computational efficiency but also complemented the shortcomings of classical algorithms when combined. The most representative example was the fusion of APF and Ant Colony Optimization (ACO) (Gao, Zhou, Zhao, & Shao, 2023), which effectively addressed the issue of APF falling into local minima (Zhuang, Sharma, Subudhi, Huang, & Wan, 2016). Similarly, the combination of bio-inspired algorithms such as Particle Swarm Optimization (PSO) (Zhuang, et al., 2016), Genetic Algorithm (GA) (Fiskin, Atik, Kisi, Nasibov, & Johansen, 2021), and classical algorithms have been successfully applied in intelligent SCA decision-making.

Table 2
The fitting formula of collision risk.

No.	References	Formula
1	(Xu & Wang, 2014)	$\lambda_i = (a\cdot dCPA_i)^2 + (b\cdot tCPA_i)^2$
2	(Hayama and Takeo, 1990)	$N_c = (a\cdot dCPA_i)^2 + (b\cdot V_c)/R$
3	(Zhen et al., 2017)	$CRI = \alpha a_d \exp(b_d DCPA) + \beta a_t \exp(b_t TCPA)$
4	(Ha, Roh, & Lee, 2021)	$CR = c\cdot\exp(-DCPA/a)\cdot\exp(-TCPA/b)\cdot F_{angle}$
5	(Im & Luong, 2019)	$PCR = f_{poiont}(p) = e^{-(\lambda DCPA^2 + TCPA^2)/2\sigma^2}$
6	(Mou, et al., 2010)	$R_{colli} = R_{basic}\exp^{-tcpa/10}\exp^{-\lvert cpa\rvert}F_{angle}$

With the advent of the Big Data era, machine learning has received unprecedented attention. For instance, in recent years, a plethora of literature has emerged using methods such as Reinforcement Learning (RL) (Kim, Park, & Cho, 2022), DRL (Xu, Lu, Liu, Cai, & Zhang, 2022), and Q-learning (Shen, et al., 2019). These methods were the hot topics in the research of intelligent SCA. They were capable of learning feature extraction from large datasets, enabling ships to truly become intelligent. The two main components of DRL are the MDP and neural networks (Zheng, Zhang, Wang, Zhang, & Cui, 2023). On one hand, neural networks express and find approximate solutions to these problems. The fast computing capability of neural network methods plays a crucial role here, as they can train, learn, and remember historical data (Wu, et al., 2023). Q-learning combined with neural networks in the form of Deep Q Networks (DQN) was the fundamental basis of DRL (Deraj, et al., 2023). Depending on the research problem, DQN has produced many variants, such as Deep Neural Network (DNN) (Zhao & Roh, 2019), Long Short-term Memory Networks (LSTM) (Gao & Shi, 2020), Convolutional Neural Network (CNN) (Zhang, Feng, Goerlandt, & Liu, 2020), Recurrent Neural Network (RNN) (Shi & Liu, 2020), and Generative Adversarial Network (GAN) (Zhang, et al., 2023). In particular, integrating DQN and LSTM, known as Deep Recurrent Q-Network (DRQN), enables intelligent ships to have memory capabilities (Jiang, et al., 2022).

On the other hand, MDP is the model/language that must be expressed for problems solved using DRL. It was commonly used to represent or model the environment component in RL (Zheng, et al., 2023). Additionally, depending on the different ways of policy optimization and value function acquisition, DRL has also produced various algorithmic variants, such as Proximal Policy Optimization (PPO) (Rongcai, Hongwei, & Kexin, 2023), Deep Deterministic Policy Gradient (DDPG) (Guo, Zhang, Zheng, & Du, 2020), and Soft Actor-Critic (SAC) (Song, Xu, Hao, Yao, & Guo, 2022). Machine learning has strong compatibility with other algorithms. It can be combined with classical algorithms, such as DRL combined with APF, which solves the problem of easy local iteration and slow convergence speed in MASS autonomous navigation decision simulations (Zhang, Wang, Liu, & Chen, 2019). Moreover, machine learning was able to mix with some bio-inspired algorithms to achieve unexpected results. For example, the improved Q-learning Beetle Swarm Antenna Search (I-Q-BSAS) algorithm takes into account ship maneuverability, collision risk, and international maritime SCA rules in multi-ship encounters (Xie, Garofano, Chu, & Negenborn, 2019). The third-generation SCA decision-making technology, which was primarily based on machine learning and bio-inspired algorithms, makes ship intelligent decision-making more anthropomorphic and intelligent (Song, Kayano, & Shoji, 2023). Fortunately, the third-generation SCA decision-making technology not only effectively addressed the complexity of ship motion environment and the uncertainty of ship behavior, but also perfectly complemented SCA geometric principles and COLREG (Rongcai, et al., 2023).

In Table 3, we conducted a comparative analysis of the main algorithms of the three generations of SCA technologies from the perspectives of algorithm principles, advantages and disadvantages, and application objects. The intelligent SCA algorithms represented by mathematical models mainly address the risk assessment and collision avoidance decision optimization problems in traditional ships. These algorithms are pioneers in intelligent SCA technology, aiming to reduce the burden on crew members and lower the accident occurrence rate. The second-generation SCA technologies, represented by intelligent

Table 3
Advantages, disadvantages, and application object of SCA algorithms.

Classification		Algorithms	Strengths	Weaknesses	Application object	References
Mathematical model		Index reasoning	More influencing factors can be included	Strong specificity	Traditional ship	(Liu, et al., 2020)
		Formula fitting	Wide coverage and fast calculation speed	Heavily dependent on parameters	Traditional ship	(Mou, et al., 2010)
		Ship domain	Can reflect the risk value from all directions	Easily influenced by the shape of the field	Traditional ship	(Liu, et al., 2022)
Intelligent algorithm		A*	In response to the environment quickly, search path directly	The real-time performance is poor and the calculation time is long	Traditional ship	(Singh, et al., 2018)
	VO	NLVO	Convert discrete time slices into continuous processes	It depends on the standard collision candidate set	Traditional ship	(Chen, et al., 2020)
		PVO	Consider an uncertain dynamic environment	Ignoring OOW's preference for SCA	Traditional ship	(Li, et al., 2022)
		RVO	Solve PPO sparse reward problem	The path planning policy has a single output	USV	(Xue, et al., 2023)
		GRVO	Solve the problem of multi-ship cooperative SCA in mixed scenarios	Constrained by communication	Traditional ship	(Wang, Zhang, Song, et al., 2023)
		APF	The potential field can be implemented by the ship field	It is easy to fall into local optimality	USV	(Yang, et al., 2011)
	Bionic Algorithms	ACO	It is not easy to fall into local optimality	The local cross path generated by the algorithm during the search process	Traditional ship	(Gao, et al., 2023)
		GA	Fast calculation time	Dependencies define fitness functions and chromosome structure	ASV	(Fiskin, et al., 2021)
		PSO	Has global search capability and	It is easy to fall into local extreme points	AUV	(Zhuang, et al., 2016)
Machine learning		RL	Applicable to autonomous ship learning environment problems	Limited by the realistic environment of ship learning	MASS	(Kim, et al., 2022)
		Q-learning	Handling complex SCA tasks with multiple ships	Limited availability for emergencies	USV	(Shen, et al., 2019)
	DRL	DQN	It is possible to acquire the ability to make optimal decisions through sufficient experience.	The problem of continuous motion space is not applicable	USV	(Deraj, et al., 2023)
		CNN	It is especially suitable for image recognition of ship collision risk	An unbiased classification of risk levels for the encounters of different numbers of ships is not possible	MASS	(Zhang, et al., 2020)
		GAN	It can accurately predict the relative position information of multi-ship in multi-ship prediction task	Rely on good training methods	MASS	(Zhang, et al., 2023)
		RNN	Suitable for processing sequence data and prediction problems	Long training time	USV	(Shi & Liu, 2020)
		DNN	The nonlinear fitting ability is very strong	Relying on large amounts of data and computing resources	MASS	(Zhao & Roh, 2019)
		LSTM	Strong learning ability	Long training time	MASS	(Gao & Shi, 2020)
		DDPG	It has a strong deep neural network function fitting ability and better generalization learning ability	It might fall into local optimality	MASS	(Guo, et al., 2020)
		SAC	Suitable for trajectory control, with strong anti-interference ability	High sensitivity to hyperparameters	USV	(Song, et al., 2022)
		PPO	Control can be output in continuous action space	Requires strong hardware support	MASS	(Rongcai, et al., 2023)
	Hybrid intelligent algorithm	DRL-APF	It avoids huge computation in complex environments	Couldn't explain the security issue	MASS	(Zhang, et al., 2019)
		I-Q-BSAS	Solve MPC optimization problems	The parameters need to be adjusted repeatedly	MASS	(Xie, et al., 2019)

algorithms, serve as a transition from traditional ships to MASS. We can observe the application of such technologies in both traditional ships and MASS. Classic algorithms such as A*, VO, and APF have high application value in route planning but tend to get trapped in local optima when facing complex navigation scenarios (Sang, You, Sun, Zhou, & Liu, 2021; Yang, et al., 2011; Zheng, et al., 2024). Under the advocacy of the IMO, autonomous ships have emerged, and the third generation of intelligent SCA algorithms is dedicated to achieving unmanned navigation of ships. Intelligent bio-inspired algorithms greatly solve the local optima problem in classic algorithms, leading to the development of hybrid intelligent algorithms (Gao, et al., 2023; Yu & Roh, 2024). Machine learning techniques can effectively handle big data environments, enabling the formulation of human-like strategies through learning from historical data (Gao & Shi, 2020). However, its accuracy is severely affected by the influence of data samples.

Currently, a significant number of experiments on intelligent SCA technology are conducted using simulation and virtual reality techniques. With the continual iteration and advancement of technology, various SCA techniques have shown promising results in USVs. For instance, Woo and Kim (2020) conducted collision avoidance experiments with USVs in various scenarios, validating the applicability and effectiveness of DRL technology considering COLREGs in real-world situations. Additionally, Wang, Yu, Liang, and Li (2018) focused on the design and validation of suboptimal obstacle avoidance algorithms for unmanned boats, confirming that Layered Navigation and LNDT can indeed guide USVs to navigate safely in the presence of static and dynamic obstacles. The most exciting development lies in the successful demonstration of the multi-ship collision avoidance system tailored for open waters during the on-site testing of the ship “Zhi Fei” This system exhibited commendable performance, indicating a significant level of reliability in practical maritime operations for intelligent SCA technology. As technology continues to advance and undergo refinement, it is anticipated that intelligent SCA technology will play an increasingly substantial role in enhancing maritime safety.

5. Discussion

5.1. Knowledge discovery in SCA study database

The development of intelligent SCA in the maritime domain is closely related to traditional SCA research. SCA research can be traced back to the 1960s and 1970s when some developed countries in the West quantified certain qualitative concepts of ship maneuvering in the early regulations (Zhu, et al., 2020). The document bibliometrics analysis in this research included only relevant literature from the past 20 years. To gain a comprehensive understanding of the development of collision avoidance research, this section provides a systematic review of relevant studies from the past several decades. An overview of the development of SCA research over the past half-century is presented in Fig. 12.

After decades of research, the evolution of intelligent SCA in the maritime field can be observed in three aspects. Initially, the focus of research shifted from traditional ships to autonomous ships as the ultimate goal, with MASS serving as a transitional phase. Subsequently, the research perspective emphasized three key aspects which are collision risk, ship behavior, and decision-making techniques. Ultimately, the research techniques have undergone three stages at the algorithm level, which are the mathematical model stage, the rule-based SCA algorithm stage, and the intelligent algorithm stage. This evolution highlights the importance of integrating traditional SCA research into the development of intelligent SCA in the maritime field. By building upon the knowledge and insights gained from decades of research, the field made significant progress toward the goal of achieving safe and efficient ship operations through intelligent SCA systems.

1) Trends in the SCA object

Traditional ships, which are crew-centered, relied on the judgments and decisions of the crew to navigate and avoid collisions. Traditional SCA methods are primarily based on visual observation and depend on the crew’s experience and skills to detect and avoid potential collisions. Nevertheless, owing to the limitations of human vision and judgment, these SCA methods are susceptible to the effects of weather, sea conditions, and navigation environments, which pose significant safety risks. Research showed that 92 % of marine accidents are caused by human error (Xi, Yang, Fang, Chen, & Wang, 2017). To overcome the limitations of traditional SCA, MASS has emerged. MASS employs various sensors and equipment to obtain more precise environmental perception and achieve autonomous navigation and SCA. The advantages of MASS lay in their ability to efficiently obtain comprehensive environmental information and autonomously determine and execute SCA decisions, greatly improving the safety and reliability of ship operations. With the application of artificial intelligence and machine learning, intelligent SCA become the mainstream direction of intelligent SCA research. By introducing various intelligent algorithms and data analysis techniques, intelligent ships could achieve real-time monitoring and analysis of ship operations, dynamically deduce optimal decisions under various scenarios, and provide decision-making support and assistance to crew members. The promotion of AIS equipment and data sharing drove the development of intelligent SCA research (Zhang, Montewka, Manderbacka, Kujala, & Hirdaris, 2021). In intelligent SCA research, path planning receives the most attention. From a research perspective, it progresses from static and dynamic obstacles, single-ship and multi-ship scenarios, open water, and complex water environments, to self-avoidance and target-ship avoidance. From the review of results, current research focuses on collision avoidance in open waters, with fewer studies on collision avoidance in narrow waters and traffic-dense areas with multiple ships. In the perception module, the credibility and completeness of the perception of dynamic and static targets in complex sea conditions are still not high enough. For example, the reliability of detecting small targets like small fishing boats and floating objects on the sea surface is low, which does not fully meet the safety requirements for autonomous ship navigation. Additionally, as the final stage of path

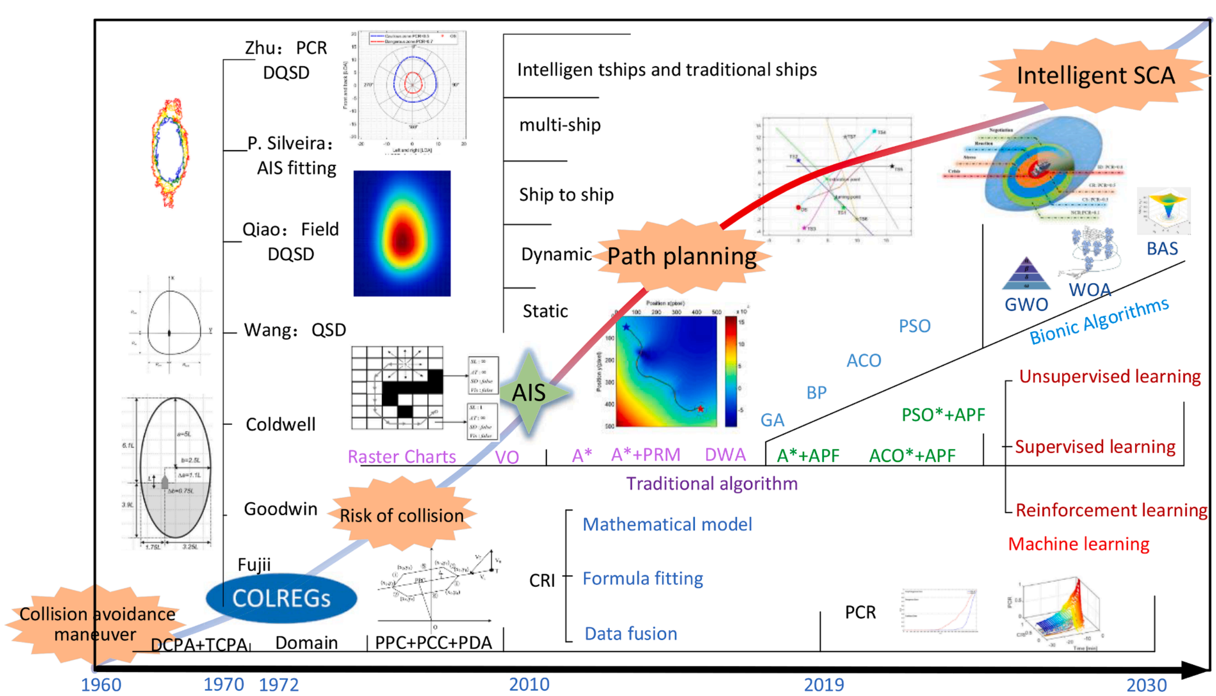

Fig. 12. Evolution of SCA studying.

planning, motion planning is considered a micro-modality constraint problem. Unlike path planning, which assumes the ship to be a point mass, motion planning deals with the rigid body having either three or six degrees of freedom (Zhang, et al., 2024). By introducing kinematics and dynamics, and employing control theory along with random sampling methods (such as PRM and RRT techniques), a refined SCA decision-making solution is derived, taking into account factors such as forces and moments of inertia. However, current stage path planning and motion planning mostly assume scenarios where the evasive objects are also intelligent ships. In scenarios involving encounters between intelligent ships and traditional ships within a multi-modal environment, it becomes necessary to consider inter-ship interaction coordination within a mixed traffic environment (Liu, et al., 2023). This is achieved through multiple rounds of interaction via the construction of inter-ship self-organizing topological coordination networks to reach consensus decisions. Additionally, despite the anthropomorphized intelligent SCA reducing information asymmetry between intelligent ships and traditional ships by adhering to COLREG, it is still necessary to consider the uncertainty of ships' behavior to enhance the collaborative ability between intelligent ships and traditional ships, thereby ensuring navigational safety (Xin, Liu, Yang, Zhang, & Wu, 2021; Zhang, et al., 2023). With the improvement of intelligent ship technology, it will be an interesting topic to consider the collision avoidance of multi-mode encounter scenarios between intelligent ships and traditional ships.

2) Trends in the SCA technology

Geometric avoidance remains fundamental in intelligent SCA. The emergence of radar plotting systems undoubtedly represents a significant achievement in developing geometric avoidance theory. Whereas, research on intelligent SCA based on radar is currently rare. From the perspective of collision risk, mathematical modeling is the mainstream approach in geometric avoidance research. DCPA and TCPA can serve as indicators of collision risk individually or collectively to assess collision danger. This ultimately leads to three main directions in collision danger assessment which are the fusion of multiple factors, formula fitting-based assessment, and integration of big data. It is worth mentioning that with increasing attention to near-miss incidents, potential collision risks and real-time collision risks have garnered significant attention. Furthermore, the introduction of collision risk in the maritime field presents another perspective. With the promulgation of the 1972 edition of the COLREG, research on SCA began to focus on these regulations. The direction in the ship domain has shifted from the original ellipse to an inclined ellipse, and finally to quaternions, all due to the existence of these regulations. In the present, the ship domain is more likely to analyze the probability density distribution of ship positions within specific areas from a big data perspective. Additionally, the ship domain should be dynamic. From the perspective of collision probability prediction, the potential collision point and potential collision area of a ship should be given more attention in intelligent SCA. Traditional SCA often treats the maritime domain as an obstacle to be avoided. Nevertheless, from the standpoint of avoidance geometry principles, the potential collision area may be more suitable as an avoidance target. Furthermore, ships in encounter situations often exhibit information asymmetry, individual variability, and target coordination. It is essential to focus on the behavioral intentions of the target ship to enhance the coordination and real-time optimization of collision avoidance decision-making.

3) Trends in the SCA algorithm

Intelligent SCA decision-making in ship navigation has revolutionized traditional SCA principles. The technological advancements in SCA decision-making have evolved through iterative updates of intelligent algorithms to adapt to the complexity, uncertainty, and big data ecosystem of the maritime environment. From the perspective of intelligent algorithms, this progression can be divided into four stages: classical intelligent algorithms, biomimetic intelligent algorithms, hybrid intelligent algorithms, and machine learning.

In the early stage, algorithms such as A*, APF, and RRT (Enevoldsen, Reinartz, & Galeazzi, 2021), and dynamically feasible path planning only addressed global or local path planning problems. The emergence of biomimetic algorithms improved optimization efficiency and enhanced intelligent SCA reliability. Hybrid intelligent algorithms can address multiple issues in intelligent SCA while ensuring reliability and stability. Looking ahead, the complex encounter scenarios in the maritime ecosystem necessitate a shift from single intelligent algorithms to multimodal techniques. Importantly, SCA rules remain applicable in intelligent SCA, representing both a challenge and a key factor in aligning intelligent SCA with practical maritime situations. Since the pioneering work of Deraj, et al. (2023) in applying the DQN method to SCA, Reinforcement Learning approaches have gained widespread attention in the field due to their simplicity and adaptability. The traditional algorithms for SCA decision-making are still unable to effectively deal with complex encounter situations in narrow and intensive waterways. The avoidance behaviors of own ships are overly conservative. In many research studies, there is a tendency to adopt earlier and more drastic evasion behaviors or even "evasion strategies", resulting in unnecessary detours that impact navigation efficiency. It is urgent to develop efficient algorithms for multi-ship autonomous coordinated collision avoidance that can achieve interactive and coordinated collision avoidance between intelligent ships and traditional ships in mixed traffic scenarios, as well as autonomous collision avoidance in restricted waterways. Deep learning is widely recognized as a superior technology for addressing complex scenario problems. The advancement of big data has accelerated the development of both reinforcement learning and deep learning. Currently, the application of deep learning in intelligent SCA primarily focuses on three topics: one involves issues of uncertain ship quantities and risk magnitudes in complex scenarios. Another pertains to SCA problems involving COLREG (Wang, et al., 2024). The third topic concerns trajectory prediction problems based on uncertain ship behaviors. These implies that SCA is increasingly trending towards anthropomorphism and intelligence, necessitating the integration of more technologies to address different types of SCA problems. Given this context, the future trend of intelligent SCA will primarily rely on reinforcement learning and deep learning, integrating traditional intelligent algorithms and biomimetic algorithms from various perspectives, to maximize the realization of the "navigation brain" in the SCA process. This integration represents a crucial technological advancement in intelligent SCA research.

5.2. Contributions and limitations of this review

The contribution of this article is twofold. One aspect of the contribution involves summarizing research guidelines in the field of intelligent SCA. Another aspect is manifested in the methodology, where a novel technique called the literature portrait technique is applied in the literature review process. With the rapid increase in literature across various fields, conditions have been created for bibliometric analysis. Nevertheless, there has been little application of bibliometrics in literature reviews of the SCA industry. According to the statistics presented in Table 4, this paper elucidates the advantages of using bibliometric portrait techniques to review literature for SCA, as compared to the literature reviews of the past 15 years. Initially, bibliometric portrait techniques are capable of covering a greater number of literature than traditional statistical methods. Previously, in contrast to bibliometric methods, bibliometric portrait techniques not only possess the same functions, but also provide accurate flow relationships among organizations, authors, and research hotspots, as well as the close collaborative relationships among authors. Eventually, the utilization of bibliometric portrait techniques provides a new perspective and an in-depth review of SCA research. This method allows for greater attention to be paid to various aspects of the SCA field in a more precise and comprehensive manner. Before this, most existing reviews were typically conducted through manual analysis and categorization of the chosen literature and discussions on research topics and trends.

It is imperative to consider the following limitations when

Table 4
Comparison between this study and the relevant SCA review studies.

No.	References	Documents	Methods					Contributions
			Classification	Review	bibliometrics	Statistics	portrait	
1	(Tam, Bucknall, & Greig, 2009)	57	√	√				This study provides a comprehensive review of the development of SCA techniques and path planning in the context of maritime ship navigation
2	(Liu, et al., 2022)	104	√	√	√			It specifically focuses on the development and trends of MASS.
3	(Akdag, et al., 2022)	288	√	√				In contrast to previous discussions on SCA algorithms, this study distinguishes itself by considering and emphasizing the importance of collaboration among relevant ships.
4	(Ozturk & Cicek, 2019)	64	√	√		√		To thoroughly evaluate and analyze the existing literature on this topic, this study employs two dimensions: model-based analysis and parameter clarification.
5	(Lazarowska, 2021)	66	√	√				It compares and analyzes the latest SCA algorithms and real-time path planning algorithms.
6	(Campbell, Naeem, & Irwin, 2012)	80	√	√				Building upon notable research conducted thus far, which primarily encompasses navigation, guidance, control, and motion planning, the study addresses specific development requirements.
7	(Zhou, et al., 2020)	114	√	√		√		The study also introduces the research progress in path planning based on multimodal constraints.
8	(Öztürk, Akdag, & Ayabakan, 2022)	117	√	√		√		It explores the path planning algorithms for autonomous maritime vehicles and their relevance to collision control.
9	(Lexau, Breivik, & Lekkas, 2023)	231	√	√		√		Automated Docking for Marine Surface Vessels-A Survey
10	(Wrobel, Gil, Huang, & Wawruch, 2022)	113	√	√		√		The Vagueness of COLREG versus Collision Avoidance Techniques-A Discussion on the Current State and Future Challenges Concerning the Operation of Autonomous Ships
11	(Vagale, Oucheikh, Bye, Osen, & Fossen, 2021)	67	√	√				Path planning and collision avoidance for autonomous surface vehicles I: a review
12	(Vagale, Bye, Oucheikh, Osen, & Fossen, 2021)	45	√	√		√		Path planning and collision avoidance for autonomous surface vehicles II: a comparative study of algorithms
13	This Research	526	√	√	√	√	√	The document portrait technique is put forward to obtain the research trend of SCA from three angles which are object, technique, and algorithm

interpreting the results of this study. In the beginning, in previous research, Wang, Yan, Ma, Wu, and Liu (2022) proposed the concept of the navigational brain, which suggests that intelligent SCA consists of four steps: risk perception, risk identification, avoidance decision-making, and avoidance control. Conversely, this study focuses more on the issues related to intelligent SCA decision-making, neglecting the other three steps, particularly the problem of intelligent SCA control. Even though the study takes into account the collision risk involved in the steps of risk perception and identification, as well as the domain of ship operations. Subsequently, our research objects were chosen based on thematic terms. There might have been omissions in the study, where the thematic terms used might have differed from those discussed in this study, but their research findings were crucial for intelligent SCA research. Last but not least, result of the limitations in our understanding of the databases, this study had solely relied on the WoS database. Furthermore, the database might have incompleteness in its literature coverage, potentially unable to encompass all relevant literature in this domain. Moreover, for practical considerations, manual screening was applied to the database literature to obtain more representative sources. This might have resulted in the possibility that the literature included in this analysis might still not have comprehensively covered all the literature in the field of intelligent SCA. Consequently, while the findings of that study provided valuable insights into the research trends, caution had to be exercised to account for these limitations and consider the broader context of intelligent SCA research.

6. Conclusion

This research has comprehensively reviewed SCA in maritime field to understand the current trends and frontiers of intelligence SCA. A comprehensive review of nearly 526 papers published between 2004 and 2023 has been conducted, and literature profiling analysis has been used to determine the collaborative network of relevant journals, authors, countries/regions, and institutions in the field over the past 20 years. The study has traced the history of SCA research back to the 1960s and 1970s and has comprehensively introduced the research hotspots and development trends of the field of intelligence SCA from the perspectives of research objects, technical methods, and SCA algorithms.

Researchers in maritime field will focus on intelligent SCA between intelligent ships and traditional ships in complex multi-ship encounter situations from the perspective of research objects. In terms of technical methods, attention will be paid to real-time and potential collision risks based on big data, dynamic ship domains, and humanized intelligent SCA decisions. At the algorithm level, geometric SCA algorithms will remain the foundation, but they will be iterated by advanced algorithms such as deep learning. Multi-modal hybrid algorithms will also be favored. The in-depth research of smart ship collision avoidance technology can improve the safety and efficiency of maritime traffic. At last, this study compares with similar literature reviews, highlighting the advantages of literature portraits in literature reviews, and demonstrating the contribution of this study to the field of intelligence SCA.

CRediT authorship contribution statement

Qinghua Zhu: Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Yongtao Xi:** Funding acquisition, Methodology. **Jinxian Weng:** Software, Validation, Visualization. **Bing Han:** Investigation, Writing – review & editing. **Shenping Hu:** Funding acquisition, Investigation. **Ying-En Ge:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China under Grant No. 52272353.

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